

An integral of hyperbolic tangents: residues, characters and the conductors that occur

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Abstract

For $n \geq 2$ let $I_n = \int_0^\infty x^{-2} \tanh(x) \tanh(2x) \cdots \tanh(nx) dx$. On Mathematics Stack Exchange it was observed that for odd n the value is a rational combination of logarithms and values $L'(-1, \psi)$ of even Dirichlet characters, with coefficients and conductors that look patternless, and asked for a general reduction. We give one. The poles of $\prod_{k \leq n} (1 - q^k)/(1 + q^k)$ are simple, at the primitive $2m$ -th roots of unity with $\lfloor n/m \rfloor$ odd, and each residue is an explicit constant times a product of ρ cotangents, where $\rho = n \bmod m$. Expanding the residues in characters gives every coefficient of $L'(-1, \psi)$ as a finite character sum. Because a product of ρ of these cotangents equals $\pm$ the product of $m - 1 - \rho$ of them, only $\rho' = \min(\rho, m - 1 - \rho)$ matters, and families with $\rho' = 0$ bring no new conductor; this accounts for every conductor that appears or disappears in the observed lists. The finite trigonometric identities behind the residues are checked in the Lean proof assistant.

1 The question

Put $q = e^{-2x}$. Since $\tanh(kx) = (1 - q^k)/(1 + q^k)$,

$$I_n = \int_0^\infty \frac{P_n(e^{-2x})}{x^2} dx, \quad P_n(q) = \prod_{k=1}^n \frac{1 - q^k}{1 + q^k} = 1 + \sum_{N \geq 1} c_n(N) q^N.$$

The integrand behaves like $n! x^{n-2}$ at 0 and like x^{-2} at infinity, so I_n converges for $n \geq 2$. The question [1] records that $I_n = -2D'_n(-1)$ for the Dirichlet series $D_n(s) = \sum_{N \geq 1} c_n(N) N^{-s}$, that for even n this gives Clausen values, and that for odd n it gives logarithms and $L'(-1, \psi)$ for even characters ψ :

$$\begin{aligned} I_3 &= \ln \frac{27}{4}, \\ I_5 &= 4 \ln 4 - 3 \ln 3 - 5 \ln 5 + 16 \eta'(-1) + 4 L'_8(-1), \\ I_7 &= 10 \ln 5 + 7 \ln 7 - 12 \ln 3 - \frac{68}{5} \ln 2 + 20 L'_5(-1) - 4 L'_{12}(-1), \end{aligned}$$

with L_q the even real character of conductor q . From $n = 9$ on, complex characters are needed, and the conductors that occur are, for $n = 9, 11, 13, 15$,

$$\begin{aligned} &\{1, 7, 8, 12, 16\}, & \{1, 5, 7, 8, 9, 16, 20\}, \\ &\{1, 5, 8, 9, 11, 12, 16, 20, 24\}, & \{1, 5, 7, 8, 9, 11, 12, 13, 20, 24, 28\}. \end{aligned}$$

Conductor 12 appears, disappears and returns; 16 is missing at $n = 15$ although $1 + q^8$ still divides the denominator. The question asks for the general reduction and its coefficients.

Before reading on, try the smallest case: why does I_3 contain no L -value at all?

2 Residues

The denominator $\prod(1 + q^k)$ vanishes only at roots of -1 , so every pole of P_n is a primitive $2m$ -th root of unity $\zeta = e^{i\pi j/m}$ with j odd and coprime to m . We use the angles

$$x_k = \frac{\pi j k}{2m} \quad (k \in \mathbb{Z}),$$

and $\varepsilon_m(j) = 1$ for odd m , $\varepsilon_m(j) = \tan(\pi j/4) = \pm 1$ for even m (the character χ_{-4} of j).

Theorem 1. *P_n has a pole at $\zeta = e^{i\pi j/m}$ exactly when $\lfloor n/m \rfloor$ is odd, and the pole is simple. Write $n = (2e + 1)m + \rho$ with $0 \leq \rho < m$. Then*

$$a_m(j) := \lim_{q \rightarrow \zeta} \left(1 - \frac{q}{\zeta}\right) P_n(q) = \frac{2}{m} \cdot \frac{4^e e!^2}{(2e + 1)!} \cdot (-i)^{m-1} \varepsilon_m(j) \prod_{k=1}^{\rho} i \cot x_k,$$

and $c_n(N) = \sum a_m(j) \zeta^{-N}$ for every $N \geq 1$, the sum running over all poles.

Proof. Put $q = \zeta u$ and let $u \rightarrow 1$. The factor k has a pole when $\zeta^k = -1$, that is when k is an odd multiple of m , and a zero when $\zeta^k = 1$, when k is a multiple of $2m$. With $t = \lfloor n/m \rfloor$ there are $\lceil t/2 \rceil$ factors of the first kind and $\lfloor t/2 \rfloor$ of the second, so the order of the pole is $t \bmod 2$. For $t = 2e + 1$, a factor of the first kind is $(1 + u^k)/(1 - u^k) \sim 2/(k(1 - u))$ and one of the second is $(1 - u^k)/(1 + u^k) \sim k(1 - u)/2$. Over $k = (2i + 1)m$, $i \leq e$, and $k = 2im$, $1 \leq i \leq e$, these give $(2/m) 4^e e!^2 / (2e + 1)!$ times $1/(1 - u)$.

Every other factor has $\zeta^k \neq \pm 1$ and tends to $(1 - e^{2ix_k})/(1 + e^{2ix_k}) = -i \tan x_k$. Their product is computed by pairing angles: $\tan x_{m-k} = \cot x_k$ because j is odd, and $\tan x_{2m-k} = -\tan x_k$. The first pairing gives $\prod_{k=1}^{m-1} \tan x_k = \varepsilon_m(j)$, the middle factor $\tan(\pi j/4)$ surviving when m is even. The second shows that each full period $2mt < k < 2m(t + 1)$, $k \neq (2t + 1)m$, contributes $\prod_{k=1}^{m-1} (-i \tan x_k)(-i \tan x_{2m-k}) = \prod_{k=1}^{m-1} \tan^2 x_k = \varepsilon_m(j)^2 = 1$. What remains is $k = 2em + r$ with $1 \leq r < m$, which gives $(-i)^{m-1} \varepsilon_m(j)$, and $k = 2em + m + r$ with $1 \leq r \leq \rho$, where $\tan x_{m+r} = -\cot x_r$ gives $i \cot x_r$.

Since P_n has the same degree in numerator and denominator and only simple poles, $P_n(q) = A + \sum a_m(j)/(1 - q/\zeta)$ for a constant A , and expanding each term as a geometric series gives the coefficients. $\square$

Figure 1 shows the poles for $n = 9$. The constant and the tangent identities, the assembled product over all $k \leq n$, and the inversion rule below are proved in Lean; the limits of single factors are the elementary $(1 - u^k)/(1 - u) \rightarrow k$.

Replacing q by $1/q$ changes the sign of every factor, so $P_n(1/q) = (-1)^n P_n(q)$. Comparing the residues at ζ and $\bar{\zeta}$ gives $a_m(-j) = (-1)^{n+1} a_m(j)$. So for odd n the function $j \mapsto a_m(j)$ is even, and only even characters can occur, as the question observed.

3 Characters

Expand each residue family in characters modulo $2m$: $a_m(j) = \sum_{\chi} \alpha_{m,\chi} \chi(j)$ with $\alpha_{m,\chi} = \varphi(2m)^{-1} \sum_j \bar{\chi}(j) a_m(j)$. Then $c_n(N)$ becomes a combination of the sums $\sum_j \chi(j) e^{-2\pi i j N / 2m}$, and these are Gauss sums of possibly imprimitive characters. Write $\tau(\psi)$ for the Gauss sum of a primitive character ψ . For primitive ψ modulo f and every integer N , $\sum_{a \bmod f} \psi(a) e^{2\pi i a N / f} = \bar{\psi}(N) \tau(\psi)$ [2, Theorem 9.7].

The following is a form of [2, Theorem 9.12]; we include the short proof.

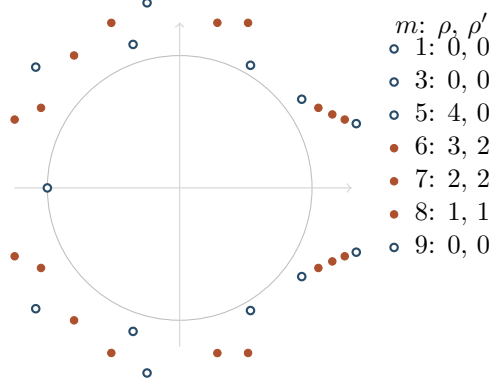


Figure 1: The poles of P_9 : the primitive $2m$ -th roots of unity for the seven m with $\lfloor 9/m \rfloor$ odd, drawn on slightly enlarged circles to separate the families. Filled points belong to families with $\rho' \geq 1$, which are the only ones that can bring a conductor other than 1; hollow points have $\rho' = 0$.

Lemma 2. Let χ modulo Q be induced by the primitive χ^* modulo f , and $g = Q/f$. For every integer N ,

$$\sum_{\substack{j \bmod Q \\ (j, Q)=1}} \chi(j) e^{-2\pi i j N/Q} = \chi^*(-1) \tau(\chi^*) \sum_{h|(N, g)} h \mu(g/h) \chi^*(g/h) \overline{\chi^*}(N/h).$$

Proof. Write the condition $(j, Q) = 1$ as $\sum_{d|(j, Q)} \mu(d)$ and $\chi(j) = \chi^*(j)$ for $(j, Q) = 1$. A term with $\mu(d)\chi^*(d) \neq 0$ has d squarefree and coprime to f , so f divides Q/d , and the sum becomes $\sum_{d|Q} \mu(d)\chi^*(d) \sum_{b \bmod Q/d} \chi^*(b) e^{-2\pi i b N/(Q/d)}$. With $Q/d = fh$, the inner sum over $b = a + fc$ ($a \bmod f$, $c \bmod h$) is $h \overline{\chi^*}(-N/h) \tau(\chi^*)$ when $h \mid N$ and 0 otherwise, by the primitive case. Since $d = g/h$ and $\overline{\chi^*}(-1) = \chi^*(-1)$, this is the formula. $\square$

Summing over N turns the inner sum into $\sum_{h|g} \mu(g/h) \chi^*(g/h) h^{1-s} \cdot L(s, \overline{\chi^*})$. So, writing $\psi = \overline{\chi^*}$ and $H_{g, \psi}(s) = \sum_{h|g} \mu(g/h) \bar{\psi}(g/h) h^{1-s}$,

$$D_n(s) = \sum_{m, \chi} \alpha_{m, \chi} \chi^*(-1) \tau(\chi^*) H_{g, \psi}(s) L(s, \psi).$$

Differentiating at $s = -1$ gives the reduction.

Theorem 3. For odd n ,

$$I_n = \sum_{\psi} K_{\psi} L'(-1, \psi) + \sum_p \lambda_p \ln p,$$

over even primitive characters ψ and primes p , where for ψ of conductor f

$$K_{\psi} = -2 \psi(-1) \tau(\bar{\psi}) \sum_{\substack{m: \lfloor n/m \rfloor \text{ odd} \\ f|2m}} \alpha_{m, \bar{\psi}} g^2 \prod_{p|g} \left(1 - \frac{\bar{\psi}(p)}{p^2}\right), \quad g = \frac{2m}{f},$$

and the logarithms come from $H'_{g, \psi}(-1) = -\sum_{h|g} \mu(g/h) \bar{\psi}(g/h) h^2 \ln h$, each multiplied by $-2\alpha \chi^*(-1) \tau(\chi^*) L(-1, \psi)$.

The coefficients reproduce the question's closed forms term by term. At $n = 5$ they are $K_8 = 4$ and $K_1 = -48$ (the $16 \eta'(-1)$ unfolds into $-48 \zeta'(-1)$ and a logarithm); at $n = 7$, $K_5 = 20$, $K_{12} = -4$ and $K_1 = 0$; at $n = 9$, $K_7 = -20 \mp 4\sqrt{3}i$ and $K_{16} = 2 \mp 2i$ for the two conjugate characters of each conductor. For $n \leq 11$ the formula agrees with direct quadrature of I_n to 10^{-27} , and in every case up to $n = 13$ the logarithm coefficients are rational, for instance $\frac{208}{21} \ln 2$ at $n = 9$.

4 Which conductors occur

The coefficients look patternless because $\alpha_{m,\chi}$ is the character transform of a product of cotangents, and such sums have no simple closed form in general. But there is a symmetry that removes much of the apparent disorder.

Corollary 4. *For $0 \leq \rho < m$, $\prod_{k=1}^{\rho} \cot x_k = \varepsilon_m(j) \prod_{k=1}^{m-1-\rho} \cot x_k$. Hence $a_m(j)$ depends on ρ only through $\rho' = \min(\rho, m-1-\rho)$, up to a factor $\varepsilon_m(j)$ and a constant. If $\rho' = 0$, then $a_m(j)$ is a constant multiple of $\varepsilon_m(j)$ or a constant, and for odd n the family contributes only to conductor 1.*

Proof. The full product $\prod_{k=1}^{m-1} \cot x_k$ is $\varepsilon_m(j)^{-1} = \varepsilon_m(j)$, and the tail $\prod_{k=\rho+1}^{m-1} \cot x_k$ equals $\prod_{k=1}^{m-1-\rho} \tan x_k$ by $\cot x_{m-k} = \tan x_k$. For $\rho' = 0$, $\rho = 0$ leaves $\varepsilon_m(j)$ and $\rho = m-1$ leaves $\varepsilon_m(j)^2 = 1$; for odd n the first case has odd m , where $\varepsilon_m = 1$. $\square$

So a conductor $f > 1$ can appear at odd n only through a family with $f \mid 2m$ and $\rho' \geq 1$. Table 1 lists the families, and the rule accounts for every case the question singles out.

n	families m with $\lfloor n/m \rfloor$ odd, as $m(\rho, \rho')$; those with $\rho' \geq 1$ in bold
9	1(0, 0), 3(0, 0), 5(4, 0), 6(3, 2) , 7(2, 2) , 8(1, 1) , 9(0, 0)
11	1(0, 0), 2(1, 0), 3(2, 0), 6(5, 0), 7(4, 2) , 8(3, 3) , 9(2, 2) , 10(1, 1) , 11(0, 0)
13	1(0, 0), 4(1, 1) , 7(6, 0), 8(5, 2) , 9(4, 4) , 10(3, 3) , 11(2, 2) , 12(1, 1) , 13(0, 0)
15	1(0, 0), 2(1, 0), 3(0, 0), 4(3, 0), 5(0, 0), 8(7, 0), 9(6, 2) , 10(5, 4) , 11(4, 4) , 12(3, 3) , 13(2, 2) , 14(1, 1) , 15(0, 0)

Table 1: The pole families for odd n from 9 to 15. A conductor $f > 1$ can only come from a bold family with $f \mid 2m$.

- Conductor 16 needs $16 \mid 2m$, so $m = 8$. At $n = 9, 11, 13$ that family has $\rho' = 1, 3, 2$; at $n = 15$ it has $\rho = 7 = m - 1$, so $\rho' = 0$ and 16 disappears, although $1 + q^8$ is still in the denominator.
- Conductor 7 needs $m = 7$ or 14. At $n = 13$ the family $m = 7$ has $\rho = 6 = m - 1$ and $m = 14$ is not a family, so 7 is absent; at $n = 15$ it returns through $m = 14$ with $\rho' = 1$, together with 28.
- Conductor 12 needs $m = 6$ or 12. At $n = 11$ the family $m = 6$ has $\rho = 5 = m - 1$, so 12 is absent; at $n = 13$ it returns through $m = 12$.
- At $n = 3$ the families are $m = 1, 2, 3$, all with $\rho' = 0$, so only conductor 1 can occur; its coefficient vanishes, and I_3 is a single logarithm.

For every odd $n \leq 15$ the conductors predicted by Theorem 3 are exactly those of the question (which lists conductor 1 in every case; its coefficient is 0 at $n = 3$ and 7).

5 The form with $L'(2)/L(2)$

The question also writes I_5 and I_7 with $L'(2, \psi)/L(2, \psi)$ in place of $L'(-1, \psi)$, without any Euler constant or $\ln 2\pi$. For an even primitive ψ of conductor f , the functional equation [2, Corollary 10.9] gives

$$\frac{L'}{L}(-1, \psi) = -\frac{L'}{L}(2, \bar{\psi}) - \ln f + c, \quad c = \ln 2\pi - 1 + \gamma,$$

with the same c for every ψ ($L(-1, \psi) \neq 0$ since ψ is even). Multiplying by $K_\psi L(-1, \psi)$ and summing, the constant c comes with $\sum_\psi K_\psi L(-1, \psi) = -2D_n(-1)$, which is 0: the Mellin transform $\int_0^\infty x^{s-1} P_n(e^{-2x}) dx$ equals $\Gamma(s)2^{-s}D_n(s)$ and is finite at $s = -1$, where Γ has a pole [1]. So the weight of $L'(2, \bar{\psi})/L(2, \bar{\psi})$ is $-K_\psi L(-1, \psi)$. It is ± 4 at $n = 5$ and ± 8 at $n = 7$, but not an integer in general: $\frac{40}{3}$ for the trivial character at $n = 11$, and $-40 \mp 8/\sqrt{5}$ for the four characters of conductor 11 at $n = 13$ (each value shared by a conjugate pair).

6 What is left

The reduction is complete and explicit, and the conductors are explained. What we do not have is a closed form for the character sums $\sum_j \bar{\chi}(j) \varepsilon_m(j) \prod_{k \leq \rho'} \cot(\pi j k / 2m)$ themselves. Whether they have a useful closed form, perhaps through Dedekind-type sums, is the remaining part of the question's "pattern". The question's asymptotic $I_n - I_\infty \sim (64/\pi^2) n e^{-\pi\sqrt{n}}$ is not addressed here.

Verification. The residue constant, the identity $(1 - e^{i\theta})/(1 + e^{i\theta}) = -i \tan(\theta/2)$, the three tangent pairings, $\prod_{k < m} \tan x_k = \varepsilon_m(j)$, the full-period product, the assembled product over all $k \leq n$ in Theorem 1, Corollary 4 and the inversion rule are proved in Lean 4 with Mathlib [3], using only the standard axioms. The limits of single factors are elementary, Lemma 2 is a form of [2, Theorem 9.12], the Mellin relation is the question's and the functional equation is [2, Corollary 10.9]; all of them are also checked numerically. Python scripts compare the residue formula with the exact power series for $n \leq 15$, the lemma for all moduli up to 40, Theorem 3 with quadrature for $n \leq 11$ and with the question's closed forms for $n = 3, 5, 7, 9$, the conductor sets for odd $n \leq 15$, and the stated weights; deliberately broken versions are rejected. An independent program, using neither the residues nor Lemma 2, expands c_n over one period in the basis $\psi(N/h)[h \mid N]$ and recovers the conductors and coefficients for $n = 5, 7, 9$. Everything is at <https://github.com/jvjk/mathematics/tree/main/tanh-product-integral>.

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References

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